Exact sparse Wilson marginals and one step repair bounds

Research note

October 9, 2026

Abstract

We integrate one $SU(2)$ Wilson link, then any fixed family of links no two of which share a plaquette, exactly. Each integrated star contributes $e^{-6H}F(Hb)$, where b is the norm of its six-staple sum and $F(t) = 2I_1(t)/t$, with its continuous value one at zero. Extracting the conditional minimum $6-b$ leaves a bounded remainder of size $O(\log H)$. The resulting interaction is genuinely not a sum of original elementary-cell interactions, so the preceding weighted class is not closed even under one removal. Nevertheless, a retained-link edit lifts to at most seven times as many original links. The original Wilson energy moment and minimizer-cap estimate give a one-region weighted repair/exception bound in the exact marginal, with explicit local costs. For a reflection-invariant sparse pattern the complete marginal inherits site-reflection positivity; no positivity of its individual induced terms is assumed. The rough labels after integration are declared separately from missing original plaquettes. This is one exact partial marginalization, with no rescaling or repeated-RG stability, no signed covering construction, and no continuum or mass-gap conclusion. No novelty claim is made.

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1 Model and the removed-link set

We start with the *pure* Wilson model, not with an arbitrary weighted endpoint from the fourth pass. On $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^4$, $L = 2^m$, $m \geq 2$, set

$$E(U) = \sum_q s(U_q), \quad s(g) = 1 - \tfrac{1}{2}\operatorname{Tr} g, \quad \mu_H(\,\mathrm{d}U) = Z_H^{-1} e^{-HE(U)}\,\mathrm{d}U, \quad H \geq 1.$$

Every positively oriented link has a variable in SU(2); its reverse has the inverse. Each unoriented plaquette is counted once. The reference $\mathrm{d}U$ is product *probability* Haar. Let D be a fixed set of removed links such that each plaquette contains at most one link of D. Write R for the retained links and $z = U_R$. The exact marginal is $\nu_D = (U \mapsto U_R)_*\mu_H$. Selection of D from the source configuration is not part of this statement.

For $e \in D$, its six plaquettes have energy

$$E_e(u_e, z) = \sum_{j=1}^{6} \left(1 - \tfrac{1}{2}\operatorname{Tr}(u_e S_{ej}(z)^{-1})\right),$$

where S_{ej} is the three-link detour from the source of e to its target. All detour links are retained. Identify these matrices with unit quaternions, put

$$B_e = \sum_{j=1}^{6} s_{ej}, \qquad b_e = |B_e| \in [0, 6], \qquad m_e = 6 - b_e. \tag{1}$$

The quaternion inner product gives $E_e = 6 - u_e \cdot B_e$ and $m_e = \min_{u_e} E_e$. For $b_e = 0$ every u_e is a minimizer. Gauge transformations act by common left/right endpoint multiplication on the staples, preserving b_e and m_e.

Let $E_0(z)$ sum exactly those original plaquettes containing no removed link. Since the removed stars have disjoint plaquette sets,

$$E(U_D, z) = E_0(z) + \sum_{e \in D} E_e(u_e, z). \tag{2}$$

Stars may share retained links and vertices. Conditional factorization in the removed variables does not mean independence of the induced factors under the marginal law.

2 Exact integral and a uniform logarithmic remainder

Define for $t \geq 0$

$$F(t) = \frac{2}{\pi}\int_0^\pi e^{t\cos\theta}\sin^2\theta\,\mathrm{d}\theta = \sum_{j=0}^{\infty}\frac{(t^2/4)^j}{j!(j+1)!}. \tag{3}$$

Thus $F(0) = 1$ and $F(t) = 2I_1(t)/t$ for $t > 0$. This normalization agrees with the primary integral and power-series definitions [1, 10.32.2, 10.25.2]. The series can also be obtained directly by expanding the exponential and integrating the even spherical moments.

Theorem 2.1 (Exact retained density)**.** *For every admissible fixed D,*

$$\nu_D(\,\mathrm{d}z) = \frac{1}{Z_H} e^{-HE_0(z)} \prod_{e \in D}\left[e^{-6H}F(Hb_e(z))\right]\,\mathrm{d}z. \tag{4}$$

There is no extra gauge-volume factor or unknown normalizing constant: integrating the displayed numerator gives exactly Z_H.

Proof. For $b_e > 0$, left multiplication by the unit quaternion B_e/b_e rotates the linear form $u_e \cdot B_e$ to $b_e \cos\theta$. Probability Haar on $\mathrm{SU}(2) \simeq S^3$ has angular density $(2/\pi)\sin^2\theta\,\mathrm{d}\theta$. The integral over u_e is therefore $e^{-6H}F(Hb_e)$. At $b_e = 0$ it is e^{-6H}, in agreement with $F(0) = 1$. Apply Fubini to (2). All factors are positive, bounded and measurable in this finite volume. □

Set

$$r(t) = t - \log F(t), \qquad r_H = \tfrac{1}{2} + \log 24 + \tfrac{3}{2}\log(1+6H). \tag{5}$$

Lemma 2.2 (Uniform remainder including zero staple sum)**.** *For every $t \geq 0$,*

$$0 \leq r(t) \leq \tfrac{1}{2} + \log 24 + \tfrac{3}{2}\log(1+t), \quad r(0) = 0. \tag{6}$$

In particular $0 \leq r(Hb_e) \leq r_H$. For $t > 0$, $0 < F'(t)/F(t) < 1$, so $0 < r'(t) < 1$.

Proof. Write $e^{-r(t)} = \mathbb{E}_{\mathrm{Haar}} e^{-t(1-\cos\theta)} \leq 1$. On the cap $\theta \leq \rho = (1+t)^{-1/2} \leq 1$, $t(1-\cos\theta) \leq t\rho^2/2 \leq 1/2$. The previously proved elementary Haar-cap bound gives this cap mass at least $\rho^3/24$. Hence $e^{-r(t)} \geq e^{-1/2}/[24(1+t)^{3/2}]$, proving (6). The logarithmic derivative F'/F is the expectation of $\cos\theta$ under its positive tilted angular law. It is strictly less than one. Pairing angles θ and $\pi - \theta$ shows $F'(t) > 0$ for $t > 0$. □

Define

$$\mathcal{E}_D(z) = E_0(z) + \sum_{e \in D} m_e(z) = \min_{U_D} E(U_D, z), \qquad W_D(z) = -\sum_{e \in D} r(Hb_e(z)). \tag{7}$$

Then the exact same density is

$$\nu_D(\,\mathrm{d}z) = Z_H^{-1} e^{-H\mathcal{E}_D(z) + W_D(z)}\,\mathrm{d}z. \tag{8}$$

This separates an order-H leading action from a logarithmic remainder; it does not drop the boundary-forced minimum as a scalar.

Why the separation matters. For one actual frozen star, both $b = 0$ (three $+I$ and three $-I$ staples) and $b = 6$ (identical staples) are realizable. The log-integral factor $-6H + \log F(Hb)$ thus has oscillation

$$\log F(6H) = 6H - r(6H) \geq 6H - r_H.$$

It is not an $O(\log H)$ perturbation of E_0 uniformly in all boundary data. Only the remainder after extracting $6 - b$ has that bound. Moreover $r(H|B|)$ and $6 - |B|$ separately have a cusp at $B = 0$; the full expression does not. Indeed $F(t) = 1 + t^2/8 + O(t^4)$, so $-H(6-b) - r(Hb) = -6H + \log F(Hb)$ is smooth in the four coordinates of B. A Hessian argument must not differentiate the separated leading minimum as if it were everywhere smooth.

3 Support spreading and inherited reflection positivity

Proposition 3.1 (Failure of elementary-cell closure)**.** *For $H > 0$, even the marginal from one removed link generally cannot be written, up to a constant, as the exponential of a sum of functions supported on individual original elementary four-cells of retained links. This statement concerns the original cells and no new auxiliary variables.*

Proof. Take $e = (0, e_0)$. Set retained links to identity except the two parallel middle links of its $+e_1$ and $-e_1$ staples, with values $\exp(i\alpha\sigma_3)$ and $\exp(i\beta\sigma_3)$ respectively. These two links lie at transverse coordinates $+1$ and -1; no original unit cell contains both, including for period four. The other four staples are identity. In quaternion coordinates,

$$b(\alpha, \beta) = |4 + e^{i\alpha} + e^{i\beta}|, \quad b^2 = 18 + 8\cos\alpha + 8\cos\beta + 2\cos(\alpha - \beta).$$

At the origin $b = 6$, its first derivatives vanish, and $\partial_\alpha \partial_\beta b = 1/6$. Consequently

$$\partial_\alpha \partial_\beta \log F(Hb)|_{0,0} = \frac{H}{6} \frac{F'(6H)}{F(6H)} > 0. \tag{9}$$

Every cell-supported summand, and every retained original plaquette, depends on at most one of these two variables after the others are fixed. Its mixed four-point difference is zero. The smooth exact log-density has a nonzero mixed difference for sufficiently small equal nonzero angles by (9). This argument does not assume differentiability of a putative cell decomposition. If equality is asserted only Haar-almost everywhere, Fubini makes its mixed four-point difference zero almost everywhere in the common exterior and four link values. The nonzero difference of the continuous exact log-density persists on open neighborhoods of the displayed choices, which have positive Haar measure. Thus an almost-everywhere decomposition is ruled out as well. □

Larger supports do not by themselves destroy reflection positivity of the *complete exact marginal*. If a site reflection ϑ preserves D as an unoriented link set, the retained-coordinate projection commutes with it. A bounded real retained positive-half function lifts to a fine positive-half function, so

$$\mathbb{E}_{\nu_D}[f \vartheta f] = \mathbb{E}_{\mu_H}[\widetilde{f} \vartheta \widetilde{f}] \geq 0. \tag{10}$$

This uses the fine model's site-reflection positivity proved in the preceding notes. It does not assert positivity of the individual m_e or $r(Hb_e)$ interactions, or of an arbitrary non-invariant marginal.

A concrete nonempty periodic pattern is

$$D_0 = \{(x, x + e_0) : x_1, x_2, x_3 \text{ even}, \ x_0 \text{ arbitrary}\}. \tag{11}$$

It has $V/8$ links. A plaquette involving direction zero has two parallel direction-zero links at opposite transverse parities, hence contains at most one removed link. A plaquette in the other directions contains none. Normal site reflection changes the starting x_0 by an odd shift but x_0 is unrestricted; transverse site reflections preserve even parity. Thus D_0 is invariant under every integer-site coordinate reflection. The marginal retains those positivities, translations by one in direction zero, and translations by two in the other directions. It is not a Wilson model on a rescaled complete hypercubic lattice.

4 Local closure and moment bounds in the exact marginal

Fix a nonempty $J \subset R$, with $n = |J|$. Let

$$A_D(J) = \{e \in D : F_e \cap J \neq \varnothing\}, \quad a = |A_D(J)|, \quad K = J \cup A_D(J), \quad s = |K| = n + a, \tag{12}$$

where F_e consists of the 18 retained detour links in the removed star. Since an original plaquette contains at most one removed link and each retained link belongs to six plaquettes,

$$a \leq 6n, \qquad s \leq 7n. \tag{13}$$

Keep every term of every affected star, not only its plaquettes which happen to touch J. Define the local leading energy

$$G_J(z) = \sum_{\substack{q\cap J\neq\varnothing\\ q\cap D=\varnothing}} s(z_q) + \sum_{e\in A_D(J)} m_e(z). \tag{14}$$

Then $\mathcal{E}_D - G_J$ depends only on retained links outside J. For the local remainder $W_J = -\sum_{e\in A_D(J)} r(Hb_e)$, $W_D - W_J$ also depends only on that exterior. Every J-edit obeys

$$|W_D(z) - W_D(y)| \le a r_H \le 6n r_H. \tag{15}$$

The full leading energy may change by order Hn in the exponent; that change is accounted for by its energy gain, not by (15).

For an original fine-link set K, write E_K for the sum of all original plaquettes touching K. No plaquette touching K contains a removed link outside $A_D(J)$. Thus, as a function of the remaining retained data and the removed variables u_A,

$$E_K(u_A, z) = \sum_{\substack{q\cap J\neq\varnothing\\ q\cap D=\varnothing}} s(z_q) + \sum_{e\in A_D(J)} E_e(u_e, z), \qquad G_J(z) = \min_{u_A} E_K(u_A, z). \tag{16}$$

Let $w = z_{R\setminus J}$ and

$$g_J(w) = \min_{z_J} G_J(z_J, w) = \min_{z_J, u_A} E_K(u_A, z_J, w). \tag{17}$$

Both minima are continuous by compactness. Removed variables outside $A_D(J)$ are irrelevant to these functions and cannot secretly supply frozen-boundary terms.

Recall the established original Wilson moment [2]:

$$\mathbb{E}_{\mu_H} e^{HE_K/128} \le e^{b_H|K|}, \qquad b_H = \tfrac{1}{2}a_H + \frac{\log 2}{64}, \quad a_H = \tfrac{3}{2} + \log 24 + \tfrac{3}{2}\log(8H).$$

Theorem 4.1 (Inherited local energy and boundary-minimum tails)**.** *For the exact marginal of every admissible fixed D,*

$$\mathbb{E}_{\nu_D} e^{HG_J/128} \le e^{b_H(n+a)}, \qquad \mathbb{E}_{\nu_D} e^{Hg_J/128} \le e^{b_H(n+a)}. \tag{18}$$

Consequently for $v \ge 0$,

$$\nu_D(G_J \ge v),\ \nu_D(g_J > v) \ \le\ \min\{1, e^{-Hv/128 + b_H(n+a)}\}. \tag{19}$$

The second is a tail in the actual retained exterior marginal*, not at an arbitrarily fixed boundary. The estimates need no new chessboard inequality on the decimated graph.*

Proof. For every fine configuration, $0 \le g_J \le G_J \le E_K$ by (16)–(17). Expectation of a retained-coordinate function under ν_D equals its expectation under μ_H exactly. Apply the original moment and then Markov's inequality. In particular, this proof works even if D is not reflection invariant. □

5 A retained-link sublevel kernel without differentiating a cusp

Theorem 5.1 (Retained near-minimum replacement)**.** *For every J there is a Borel gauge-covariant retained-link probability kernel, independent of the original values in J, freezing all retained links*

outside J, with conditional density on the edited coordinates relative to product probability Haar on $\mathrm{SU}(2)^J$ *at most*

$$M_J = (24H^{3/2})^n = e^{\ell_H n}, \qquad \ell_H = \log 24 + \tfrac{3}{2}\log H, \tag{20}$$

and every output satisfies

$$G_J(y_J, w) \le g_J(w) + 12n/H. \tag{21}$$

The constant uses n, not the larger lift size $n + a$.

Proof. Use probability Haar restricted to the retained sublevel $\mathcal{B}_J(w) = \{y_J : G_J(y_J, w) \le g_J(w) + 12n/H\}$. It remains to bound its Haar volume.

For a fixed w, take any full minimizer (v_J, v_A) of the smooth fine energy E_K in (17). Hold v_A fixed and vary only $v_e \exp(tX_e)$ for $e \in J$, with $\|X_e\|_{\mathrm{op}} \le H^{-1/2}$. The first derivative of E_K at zero vanishes. As in the earlier minimizer-cap proof [3], the product rule and original four-link/six-plaquette incidences give

$$|E_K''(t)| \le 24 \sum_{e \in J} \|X_e\|_{\mathrm{op}}^2 .$$

Terms of E_K not touching J have zero derivative. Taylor's integral remainder therefore gives $E_K(v_J e^X, v_A, w) \le g_J(w) + 12n/H$. Since $G_J(v_J e^X, w) \le E_K(v_J e^X, v_A, w)$, the full product of n retained caps lies in $\mathcal{B}_J(w)$. Its Haar mass is at least $(24H^{3/2})^{-n}$. This proof differentiates only the smooth lifted energy, not the potentially nonsmooth G_J.

Continuity of G_J and g_J makes the sublevel Borel and its positive-volume normalization Borel by parameter integration. Gauge transformations carry the minimized fine configurations and the retained sublevel into their transformed counterparts, preserving Haar volume. Normalized restriction is therefore gauge covariant. No measurable choice of (v_J, v_A) is used to define the kernel. □

This kernel does not resample the removed links into the output, and is not claimed to preserve a finer conditional distribution, reflection positivity, mixed-loop holonomies, or a continuum topological charge. It is an existence result, not an implemented generic minimizer or sublevel sampler.

6 Which rough labels survive marginalization

The local leading energy is a sum over a fixed inventory of nonnegative *effective labels*:

- an original plaquette q containing no removed link, with value $s(z_q)$ and its four retained links as declared support;
- a removed-link star $e \in D$, with value $m_e(z)$ and its 18 retained detour links F_e as declared support.

These are labels of the exact minimized action, not a reassignment of independent plaquette matrices. Each original removed plaquette has ceased to be an observable of z alone. In particular, an m_e label is not the roughness of an individual eliminated plaquette.

Declare a label rough if its value exceeds $p^2/(2H)$, for $p > 0$. Let $N_p^D(J)$ count rough labels whose declared support meets J. The values of all effective labels whose declared support meets J sum exactly to G_J; the sum over only the rough ones is bounded by G_J. An unremoved plaquette is rough precisely at the original curvature threshold $p/\sqrt{H}$, but a star value ranges up to six rather than two. Thus all effective rough labels are empty at $p^2 \ge 12H$; the earlier all-plaquette cutoff $p^2 \ge 4H$ must not be applied to star labels.

For a positive integer k split the event $\mathcal{R}_{J,k} = \{N_p^D(J) \geq k\}$ at

$$\mathcal{A}_{J,k} = \mathcal{R}_{J,k} \cap \{g_J \leq kp^2/(4H)\}, \qquad \mathcal{B}_{J,k} = \{g_J > kp^2/(4H)\}. \tag{22}$$

These are events of the exact marginal and its retained boundary. If $kp^2 \geq 96n$, sources in $\mathcal{A}_{J,k}$ and every kernel output obey

$$N_p^D(J; y) \leq 3k/4, \qquad H(\mathcal{E}_D(z) - \mathcal{E}_D(y)) \geq kp^2/8. \tag{23}$$

Indeed $G_J(z) > kp^2/(2H)$ while $G_J(y) \leq kp^2/(4H) + 12n/H$; unaffected leading terms cancel. The output rough count is at most $2HG_J(y)/p^2$.

Theorem 6.1 (One-step weighted repair and exceptional-label control)**.** *Let $kp^2 \geq 96n$. For the retained sublevel kernel $\mathcal{K}_J$, set $T_{J,k}(z, \mathrm{d}y) = \mathbf{1}_{\mathcal{A}_{J,k}}(z)\mathcal{K}_J(z, \mathrm{d}y)$. As measures on the retained link space,*

$$\nu_D T_{J,k} \leq \exp[-kp^2/8 + \ell_H n + ar_H]\nu_D. \tag{24}$$

The exceptional branch in the same exact marginal *satisfies*

$$\nu_D(\mathcal{B}_{J,k}) \leq \min\{1, \exp[-kp^2/512 + b_H(n + a)]\}. \tag{25}$$

Consequently

$$\nu_D(\mathcal{R}_{J,k}) \leq \min\{1, e^{-kp^2/8+\ell_H n+ar_H} + e^{-kp^2/512+b_H(n+a)}\}. \tag{26}$$

No new unknown or restricted partition function enters these bounds.

Proof. On accepted input/output pairs, the exact density (8), the gain (23), and the residual edit bound (15) give

$$e^{-H\mathcal{E}_D(z)+W_D(z)} \leq e^{-kp^2/8+ar_H} e^{-H\mathcal{E}_D(y)+W_D(y)}.$$

Bound the retained-output kernel density by $e^{\ell_H n}$ and integrate the original z_J over probability Haar. Tonelli leaves the full normalized marginal on the output, proving (24) against every nonnegative output test function. Its total accepted mass is $\nu_D(\mathcal{A}_{J,k})$. Apply (19) at $v = kp^2/(4H)$ for (25), and then use $\mathcal{R}_{J,k} \subset \mathcal{A}_{J,k} \cup \mathcal{B}_{J,k}$. □

For useful explicit absorption it suffices to add

$$kp^2 \geq 16(\ell_H n + ar_H), \qquad kp^2 \geq 1024 b_H(n + a). \tag{27}$$

The two terms in (26) are then at most $e^{-kp^2/16}$ and $e^{-kp^2/1024}$. Since $a \leq 6n$ and $r_H, \ell_H, b_H = O(\log H)$, these are genuinely region-proportional costs for one exact step, independent of $|D|$ and the ambient volume. A direct moment estimate is stronger for source probabilities alone:

$$\nu_D(N_p^D(J) \geq k) \leq \min\{1, e^{-kp^2/256+b_H(n+a)}\}. \tag{28}$$

The split theorem's additional information is its accepted *output* domination. It neither removes all effective rough labels nor restores the old plaquette inventory.

7 Geometric counts, overlap and accepted composition

Each retained link lies in at most six effective supports: every one is witnessed by one of its six original incident plaquettes, whose removed link is either absent or unique. Every effective support has at most 18 links. Hence the adjacency graph in which retained links share an effective support has maximum degree at most $6(18-1) = 102$.

Let $C_p^D(e_0)$ be the actual component through a fixed retained root of all retained links incident to a rough effective label, with size zero if the root is absent. A component of size at least n contains a connected n-link subset J through that root. Each such link has a rough witness; one label can witness at most 18 links, so $N_p^D(J) \geq \lceil n/18 \rceil$. Counting at most $102^{2(n-1)}$ rooted connected subsets and applying (28) with $n + a \leq 7n$ gives

$$\nu_D(|C_p^D(e_0)| \geq n) \leq \min\{1, 102^{2(n-1)} e^{-n[p^2/4608 - 7b_H]}\}. \tag{29}$$

This pays source-dependent geometry in a single fixed marginal, not by conditioning on a selected support and then pretending its law is the original conditional Gibbs law.

Accepted operators satisfying $\nu_D T_i \leq c_i \nu_D$ compose by order of measures: $\nu_D T_1 \cdots T_r \leq (\prod_i c_i)\nu_D$. All gates and gains must be evaluated at the actual intermediate state. For a concrete adaptive accepted kernel, choose the *current* nonempty component $J = C_p^D(e_0)$, let $k = \lceil |J|/18 \rceil$, and accept $\mathcal{K}_J$ only on its low-minimum branch. Rejected sources contribute mass zero to this subprobability kernel. If

$$p^2 \geq 1728, \qquad d_H = p^2/144 - \ell_H - 6r_H > 2\log 102, \tag{30}$$

then counting the possible J in (24) gives

$$\nu_D T \leq c_H \nu_D, \qquad c_H = \frac{e^{-d_H}}{1 - 102^2 e^{-d_H}}, \qquad \nu_D T^r \leq c_H^r \nu_D. \tag{31}$$

Here $c_H < 1$ when $d_H > \log 10405$. This concerns repeated *repairs inside one fixed marginal*, not repeated integrations or an RG flow. A process that stays put on rejected sources is a different operator.

Retained exceptional identity branches still do not multiply as small operators: encountering the same event twice costs its probability once, not its square. For joint labels evaluated on one configuration, one available estimate is as follows. For a nonempty finite family of nonempty fixed J_i let $K_i = J_i \cup A_D(J_i)$, $K_\cup = \bigcup_i K_i$, and $M = \max_q |\{i : q \cap K_i \neq \varnothing\}|$. Then

$$\mathbb{E}_{\nu_D} \exp\left[\frac{H}{128M} \sum_i g_{J_i}\right] \leq e^{b_H |K_\cup|}. \tag{32}$$

The proof is the pointwise bound $\sum_i g_{J_i} \leq \sum_i E_{K_i} \leq M E_{K_\cup}$ under the fine lift. Even disjoint retained J_i can have overlapping lifted closures; M must be computed, not set to one. This is not a joint estimate for labels evaluated in different, adaptively changed exteriors.

8 What this exact step does not establish

Not the previous cell class. Proposition 3.1 is an actual counterexample to original cellwise additive support after one removal. Even after the order-H minimum is extracted, the remainder retains larger supports; the leading energy is a sum of minima, not a Wilson plaquette sum. The proof of the new bound does not import the fourth-pass cell theorem without checking its hypotheses: it instead uses the exact fine-to-marginal identity, the closure K, and a smooth fine minimizer.

The factorization hypothesis matters. If two removed links share a plaquette, (2) fails: the plaquette energy depends on both variables. Multiplying one-link integrals with the other removed link treated as frozen is not joint marginalization. The accompanying exact finite-group negative control exhibits this failure. Disjoint stars can still share retained variables; their effective weights are not probabilistically independent.

Not an iterated renormalization theorem. This operation does not introduce coarse links of a new lattice spacing, match a running coupling, or control the next exact marginal. The induced multi-staple factors will generally couple links chosen for a later removal even if they share no *original* plaquette. The one-step closure factor seven, residual costs and effective-label inventory cannot simply be reused at every depth. Starting from the fourth-pass weighted cell class also need not give a product integral: a cell potential can couple several otherwise plaquette-disjoint removed links.

No conditional repair of impossible boundaries. The bounds on m_e and g_J are marginal costs, not claims that arbitrary frozen holonomies can be made small. The frustrated star, finite-area loop constraints and flat winding holonomies from earlier passes remain. A low leading-energy label does not specify an all-small filling of each integrated star; a partial reduction in effective rough count is not an inventory-preserving surgery. Signed or complex covering activities, restricted normalizers and continuum-sector identifications are not introduced here.

Next concrete target. Specify a second exact elimination or a true block-link map, track the induced supports and retained defect inventory, and determine whether a positive endpoint class has depth-uniform local energy moments and edit costs. A valid obstruction or explicit growth law is as useful as a closure lemma: do not assume the one-step constant is stable. Joint retained defects and mandatory signed covering identities remain separate requirements. The fixed-physical-size defect test, cutoff comparison and nontrivial continuum reconstruction are still downstream.

9 Verification scope

The executable checks test probability-Haar normalization, the Bessel power series versus independent angular quadrature, the zero-staple endpoint and logarithmic remainder, exact integer noncommuting quaternion link algebra, sparse-pattern reflection geometry, lifted supports, conditional minima in finite-group models, and output/overlap budgets. Exact $\mathbb{Z}_2$ and quaternion-group integrations are finite-model diagnostics, not calculations of the four-dimensional SU(2) partition function or formal proofs of the analytic claims. Deliberately corrupted normalizations, lost star terms and falsely factorized shared plaquettes are negative controls. Existing checks and Lean scalar certificates are retained; no new analytic theorem is Lean-formalized. The separate audit files distinguish independent agent review from the author's checks; neither is external expert certification or a novelty assessment.

References

[1] NIST, *Digital Library of Mathematical Functions*, modified Bessel functions, equations 10.32.2 and 10.25.2, inspected October 9, 2026. Exact source locators are in `sources.md`.

[2] *Marginal boundary energy costs in the Wilson model*, third-pass local note, October 9, 2026, `../boundary_cost/boundary_cost.tex`, Theorem 1.1. The input remains the documented written theorem, not a Lean-certified analytic result.

[3] *Local link replacement with explicit boundary conditions*, second-pass local note, October 9, 2026, `../rough_region/rough_region_replacement.tex`, minimizer-cap lemma and robust one-link obstruction.

[4] *Weighted boundary costs and accepted local repairs*, fourth-pass local note, October 9, 2026, `../weighted_boundary/weighted_boundary.tex`, weighted output domination, elementary-cell class and its stated limits.